

TrES-1b

R-2794

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_260731 · created 2026-09-04T22:54:58+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461252.8582 +/- 0.0034 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.184 +/- 0.054
Transit depth (Rp/Rs)^2	3.38 +/- 1.98 [%]
Orbital Inclination (inc)	86.85 +/- 1.48
Airmass coefficient 1 (a1)	3.28 +/- 1.74
Airmass coefficient 2 (a2)	-0.74 +/- 0.44
Scatter in the residuals of the lightcurve fit is	49.96 %
Best Comparison Star	#2 - [221, 385]
Optimal Aperture	1.93
Optimal Annulus	3.0
Transit Duration (day)	0.084 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.184 $\pm$ 0.054 vs 0.1358 (deviation 0.64 σ)
Duration fit vs published	0.084 d vs 0.1042 d (deviation 0.86 σ)
Mid-time O–C	7.7 min
Depth SNR	2.5
Residual scatter	49.96 %

Light curve

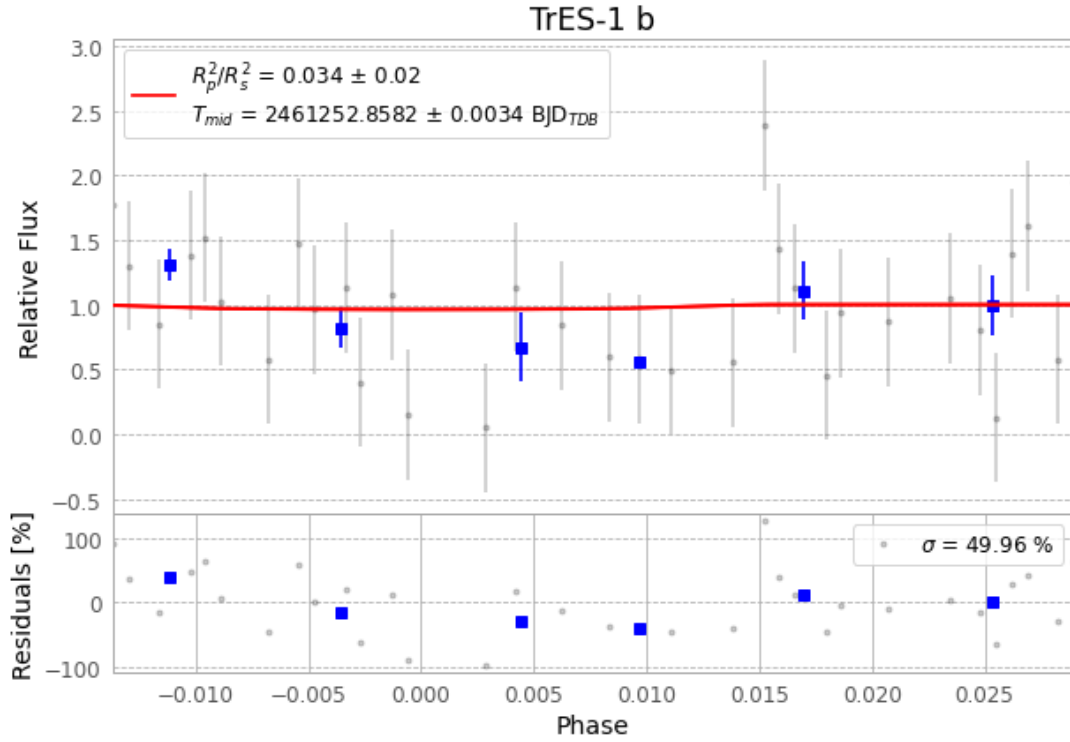


Figure 1 — FinalLightCurve_TrES-1 b_2026-07-31.png (SHA-256 1960673f1c84ad85...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [410, 165], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8807b85065622166...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

66 FITS frames (43 MB), TRES-1260731072416.FITS → TRES-1260731103915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-260731.log (da8c27e69df6...), FinalLightCurve_TrES-1 b_2026-07-31.png (1960673f1c84...), FinalLightCurve_TrES-1 b_2026-07-31.csv (dcddc66e88c6...)

Compute resources

Wall time 324.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-04 22:55Z.